

Intrinsic-Gap Control of Support Leakage for Sliced-Wasserstein Geodesics

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Lane 12, agent `agent_lane_12`, model GPT5.6

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Status. Candidate partial result, likely valid. A quantitative support-leakage bound and several complete subclasses are proved. The unrestricted ball-support question remains open in this investigation.

1 Source question

Park and Slepčev [1] define the length metric ℓ_{SW} induced by the quadratic sliced-Wasserstein distance SW on $\mathcal{P}_2(\mathbb{R}^d)$. Their Remark 4.7, on printed and PDF page 26 of arXiv:2311.05134, asks whether an ℓ_{SW} -geodesic between measures supported in the closed unit ball must remain supported there.

ing pair $(\mu_t, J_t)_{t \in [0,1]} \in \mathcal{C}\mathcal{Z}_{[0,1]}$ satisfies $\mu_t \ll J_t$, or even that J_t is a measure for a.e. $t \in [0, 1]$. See Remark 3.10. However, it can be approximated by solution of the continuity equation with $J_t \ll \mu_t$ by concatenating $(\mu_t * \eta_\varepsilon, J_t * \eta_\varepsilon)_{t \in [0,1]}$ with the Wasserstein geodesics from μ to $\mu * \eta_\varepsilon$ and $\nu * \eta_\varepsilon$ to ν , where η_ε is a suitable smooth convolution kernel with bandwidth $\varepsilon \ll 1$. $\triangle$

Remark 4.7. We note that the projection to closed balls is not a sliced Wasserstein contraction. This property holds for all transportation distances which are increasing with Euclidean distance, but we show by explicit example that it does not hold for the sliced Wasserstein distance: let

$$\mu^\varepsilon = \frac{1}{2}\delta_{(0,-1)} + \frac{1}{2}\delta_{(0,\sqrt{1+\varepsilon})}, \quad \nu^\varepsilon = \frac{1}{2}\delta_{(-\sqrt{\frac{\varepsilon}{1+\varepsilon}}, \sqrt{\frac{1}{1+\varepsilon}})} + \frac{1}{2}\delta_{(\sqrt{\frac{\varepsilon}{1+\varepsilon}}, \sqrt{\frac{1}{1+\varepsilon}})}.$$

Then $\text{supp } \nu^\varepsilon \subset \overline{B}(0, 1) \subset \mathbb{R}^2$. Let $\pi^B : \mathbb{R}^2 \rightarrow \overline{B}(0, 1)$ be the projection onto $\overline{B}(0, 1)$. Then via explicit computation one can verify that

$$SW(\pi^B_\# \mu^\varepsilon, \nu^\varepsilon) > SW(\mu^\varepsilon, \nu^\varepsilon) \text{ for sufficiently small } \varepsilon > 0.$$

In light of this observation, the following question is nontrivial: Consider $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ supported in a closed unit ball $\overline{B}$ centered at 0. Does it hold that for all $t \in [0, 1]$ the measures along the geodesic are supported within the same ball? This property sounds natural as $\hat{\mu}^\theta, \hat{\nu}^\theta$ would be supported on $[-1, 1]$ for all $\theta \in \mathbb{S}^{d-1}$, but remains an open problem. $\triangle$

Recall that in Example 2.5 established that in general the SW geodesics do not exist, whereas ℓ_{SW} geodesics between $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ always exist by Proposition 4.5. Thus we conclude this section with the following corollary.

Corollary 4.8. $(\mathcal{P}_2(\mathbb{R}^d), SW)$ is not a length space.

5. COMPARISONS WITH NEGATIVE SOBOLEV NORMS AND THE WASSERSTEIN DISTANCE

We establish two comparison results, Theorem 5.2, near absolutely continuous measures, and

Write $R_\theta \mu = (x \mapsto \theta \cdot x)_\# \mu$ and use normalized surface measure in every spherical average. Thus

$$SW^2(\mu, \nu) = \int_{\mathbb{S}^{d-1}} W_2^2(R_\theta \mu, R_\theta \nu) d\theta, \quad SW \leq \ell_{SW} \leq d^{-1/2} W_2.$$

2 Statement and proof intuition

For $\rho \in \mathcal{P}_2(\mathbb{R}^d)$ define its unit-ball leakage by

$$\mathcal{E}_B(\rho)^2 := \int_{\mathbb{S}^{d-1}} \int_{\mathbb{R}} (|r| - 1)_+^2 d(R_\theta \rho)(r) d\theta. \quad (1)$$

This quantity vanishes exactly when ρ is supported in the unit ball.

The one-dimensional W_2 quantile formula realizes SW as a chordal distance in a Hilbert space. Unit-ball support becomes a closed convex pointwise box in that space, and $\mathcal{E}_B$ is precisely distance to the box. A simple projection inequality says that a path from one point of a convex set to another cannot pass a distance r from the set with total chord length less than $\sqrt{D^2 + 4r^2}$, where D is the endpoint chord. An intrinsic geodesic has available total chord length at most its intrinsic length. The difference between these two quantities therefore pays for every possible support excursion.

Theorem 1 (Intrinsic-gap leakage bound). *Let $\mu_0, \mu_1 \in \mathcal{P}_2(\mathbb{R}^d)$ be supported in the closed unit ball, and let $(\mu_t)_{0 \leq t \leq 1}$ be any constant-speed ℓ_{SW} -geodesic between them. Set*

$$L = \ell_{\text{SW}}(\mu_0, \mu_1), \quad D = \text{SW}(\mu_0, \mu_1).$$

Then for every $t \in [0, 1]$,

$$\mathcal{E}_B(\mu_t) \leq \min \left\{ tL, (1-t)L, \frac{1}{2} \sqrt{L^2 - D^2} \right\}. \quad (2)$$

In particular, if $\ell_{\text{SW}}(\mu_0, \mu_1) = \text{SW}(\mu_0, \mu_1)$, then every ℓ_{SW} -geodesic between the endpoints stays supported in the unit ball.

3 The Hilbert-space argument

Let $Q_\mu(\theta, s)$ be the generalized quantile at $s \in (0, 1)$ of $R_\theta \mu$. The one-dimensional quantile identity gives the isometric embedding

$$\|Q_\mu - Q_\nu\|_{L^2(\mathbb{S}^{d-1} \times (0,1))} = \text{SW}(\mu, \nu). \quad (3)$$

Consider the closed convex subset of this Hilbert space

$$\mathcal{C} = \{q : |q(\theta, s)| \leq 1 \text{ for a.e. } (\theta, s)\}.$$

Pointwise clipping to $[-1, 1]$ is the metric projection onto $\mathcal{C}$. Consequently, by the quantile change of variables,

$$\text{dist}(Q_\rho, \mathcal{C})^2 = \int_{\mathbb{S}^{d-1}} \int_0^1 (|Q_\rho(\theta, s)| - 1)_+^2 ds d\theta = \mathcal{E}_B(\rho)^2. \quad (4)$$

We isolate the elementary geometry used next.

Lemma 1. *Let $\mathcal{C}$ be a closed convex subset of a Hilbert space, let $x, y \in \mathcal{C}$, and let z be arbitrary. If $D = \|x - y\|$ and $r = \text{dist}(z, \mathcal{C})$, then*

$$\|z - x\| + \|z - y\| \geq \sqrt{D^2 + 4r^2}.$$

Proof. Let p be the metric projection of z onto $\mathcal{C}$. The projection inequality $\langle z - p, q - p \rangle \leq 0$ for $q \in \mathcal{C}$ gives

$$\|z - x\|^2 \geq r^2 + \|p - x\|^2, \quad \|z - y\|^2 \geq r^2 + \|p - y\|^2.$$

Put $u = \|p - x\|$ and $v = \|p - y\|$. Since $u + v \geq D$ and $s \mapsto \sqrt{r^2 + s^2}$ is convex,

$$\begin{aligned} \|z - x\| + \|z - y\| &\geq \sqrt{r^2 + u^2} + \sqrt{r^2 + v^2} \\ &\geq 2\sqrt{r^2 + ((u + v)/2)^2} \geq \sqrt{4r^2 + D^2}. \end{aligned}$$

□

Proof of Theorem 1. Put $x = Q_{\mu_0}$, $y = Q_{\mu_1}$, and $z = Q_{\mu_t}$. The endpoint support assumption gives $x, y \in \mathcal{C}$. Since (μ_t) is a constant-speed ℓ_{SW} -geodesic and $\text{SW} \leq \ell_{\text{SW}}$,

$$\|z - x\| = \text{SW}(\mu_0, \mu_t) \leq tL, \quad \|z - y\| = \text{SW}(\mu_t, \mu_1) \leq (1 - t)L.$$

Thus each of the first two bounds in (2) follows from (4), since $x, y \in \mathcal{C}$. Their sum is at most L . Lemma 1, (3), and (4) therefore give

$$L \geq \sqrt{D^2 + 4\mathcal{E}_B(\mu_t)^2},$$

which is the final bound in (2).

It remains to justify that zero leakage is genuine support containment. Tonelli's theorem rewrites (1) as

$$\mathcal{E}_B(\rho)^2 = \int_{\mathbb{R}^d} \int_{\mathbb{S}^{d-1}} (|\theta \cdot x| - 1)_+^2 d\theta d\rho(x).$$

For every x with $|x| > 1$, the inner integral is strictly positive because an open set of directions near $x/|x|$ has $|\theta \cdot x| > 1$. Hence $\mathcal{E}_B(\rho) = 0$ forces $\rho(\overline{B}(0, 1)) = 1$. □

4 Complete subclasses

The preceding result immediately converts every equality $\ell_{\text{SW}} = \text{SW}$ into an affirmative support theorem.

Corollary 1. *Under the endpoint support hypothesis of Theorem 1, every ℓ_{SW} -geodesic remains in the unit ball in each of the following cases:*

1. *one endpoint is a Dirac mass;*
2. *both endpoints are supported on one affine line;*
3. *$\mu_1 = T_{\#}\mu_0$ for $T(x) = \lambda x + b$ with $\lambda \geq 0$;*
4. *more generally, the pair admits an extrinsic SW-geodesic.*

Proof. In the first case,

$$\text{SW}^2(\delta_a, \mu) = \int_{\mathbb{S}^{d-1}} \int |\theta \cdot (x - a)|^2 d\mu(x) d\theta = d^{-1} W_2^2(\delta_a, \mu).$$

If both measures live on an affine line with unit direction u , every projected W_2 distance is $|\theta \cdot u|$ times the one-dimensional distance along that line; spherical averaging again gives $\text{SW} = d^{-1/2} W_2$.

For $T(x) = \lambda x + b$, the map induced on every projection is the nondecreasing affine map $r \mapsto \lambda r + \theta \cdot b$, hence is optimal in one dimension. The same spherical identity gives

$$\text{SW}^2(\mu_0, T_{\#}\mu_0) = d^{-1} \int |T(x) - x|^2 d\mu_0(x) = d^{-1} W_2^2(\mu_0, T_{\#}\mu_0).$$

In all three cases $\text{SW} \leq \ell_{\text{SW}} \leq d^{-1/2} W_2$ forces $L = D$. In the fourth case the very existence of a path whose SW-length equals the endpoint SW distance likewise forces $L = D$. Apply Theorem 1. $\square$

Translations and dilations were already known extrinsic geodesics; Hopper [2] gives further extrinsic examples. Han [3] proves, under an absolute-continuity hypothesis, rigidity and stability for the different deficit $d^{-1} W_2^2 - \text{SW}^2$ and identifies affine homotheties as its equality case. Those known facts are not claimed as novel here.

5 Upgrade attempts and remaining obstruction

Metric projection. The most direct proof would project a minimizing curve radially onto the ball. Remark 4.7 of the source supplies an explicit counterexample showing that radial projection can increase SW, so endpoint contraction is false.

Quantile clipping. Pointwise clipping is contractive in the Hilbert quantile model and is exactly what produces Theorem 1. But a clipped family of one-dimensional distributions need not be the Radon transform of a probability measure. The failure of inverse Radon transforms to preserve positivity is also the source of the strict gap $\ell_{\text{SW}} > \text{SW}$ exhibited in the source paper. Thus the clipped Hilbert path cannot simply replace the original geodesic.

Particle approximation. For a curve of finitely many separated atoms, the infinitesimal sliced action is the Euclidean particle action divided by d ; Euclidean projection then does reduce length. This does not pass to arbitrary SW-absolutely continuous curves: at diffuse measures the optimal sliced flux lives in a much weaker negative-Sobolev geometry and need not be of the form $v\mu$. Approximating it by particle curves can raise the action to the Wasserstein scale.

First-exit and gap iteration. Applying the convex-set estimate at a hypothetical first exit yields the quantitative bound (2), but no contradiction when $L > D$. The strict intrinsic gap provides exactly the budget for an excursion. Neither subsegment iteration nor the endpoint moment identities eliminate that budget without an additional Radon-consistent retraction or a new local convexity theorem for the image of probability measures in quantile space.

These attempts isolate a precise obstruction. The unrestricted question is therefore not promoted to a full solution or counterexample.

6 Novelty audit and review recommendation

A bounded search on 11 August 2026 covered the run indexes; the exact source question; combinations of “intrinsic sliced Wasserstein”, “geodesic”, “support”, “ball”, “quantile”, and “length gap”; Hopper’s 2024 extrinsic-geodesic paper; and Han’s May 2026 deficit-rigidity paper. No located source states (2) or resolves the general support question. This supports, but cannot certify, novelty.

The qualitative zero-gap conclusion is close to implicit in source Remark 4.6, which characterizes a geodesic attaining SW slice by slice. Accordingly, the primary candidate-new contribution is the quantitative intrinsic-gap leakage estimate, not the known translation/dilation examples. Expert review should focus on Lemma 1 and the distance identity (4).

References

- [1] S. Park and D. Slepčev, *Geometry and analytic properties of the sliced Wasserstein space*, J. Funct. Anal. **289** (2025), article 110975; arXiv:2311.05134.
- [2] J. S. Hopper, *Sliced Wasserstein Geodesics and Equivalence Wasserstein and Sliced Wasserstein Metrics*, arXiv:2407.07219 (2024).
- [3] B.-X. Han, *Rigidity and Quantitative Stability of the Sliced Wasserstein Deficit*, arXiv:2605.25453 (2026).